

Shallow-Deep Feature Fusion for PCB Defect Detection

Improving Binary Classification Performance from 82% to 99.41%

Research Project: Semantic Aware Tiny Defect Detection in PCB using CNN

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Abstract

This report presents a novel shallow-deep feature fusion approach for binary PCB defect detection (defective vs. normal). The proposed method combines deep semantic features extracted from a pre-trained VGG16 network (512-dim) with hand-crafted shallow features (17-dim) including statistical moments, gradient characteristics, and histogram distributions. Evaluated on the DeepPCB dataset with 676 test samples, the fusion model achieved 99.41% accuracy, 99.55% precision, and 98.67% recall, representing a 17.4 percentage point improvement over the baseline VGG16 classifier (82% accuracy). Only 4 errors were observed: 1 false positive and 3 false negatives. The results demonstrate that shallow features provide complementary information that significantly enhances the model's ability to distinguish subtle defect patterns from normal manufacturing variations.

1 Introduction

Early experiments with a standard VGG16 classifier revealed a critical limitation: the model achieved only 82% accuracy with 63% defect recall, missing 37% of defective PCBs. This poor performance was attributed to the model's inability to distinguish subtle defect patterns (e.g., mouse bites, short circuits) from normal variations in copper texture, lighting, and manufacturing tolerances.

To address this limitation, we propose a **shallow-deep feature fusion** approach that combines:

1. **Deep features** (512-dim): Semantic features extracted from VGG16 pre-trained on ImageNet
2. **Shallow features** (17-dim): Hand-crafted texture features computed directly from the image patch

The fusion model learns to leverage both high-level semantic information and low-level textural cues, resulting in dramatically improved defect detection performance.

2 Methodology

2.1 Shallow Feature Extraction

For each 128x128 image patch, we extract 17 hand-crafted features:

Table 1: Shallow Feature Set (17 dimensions)

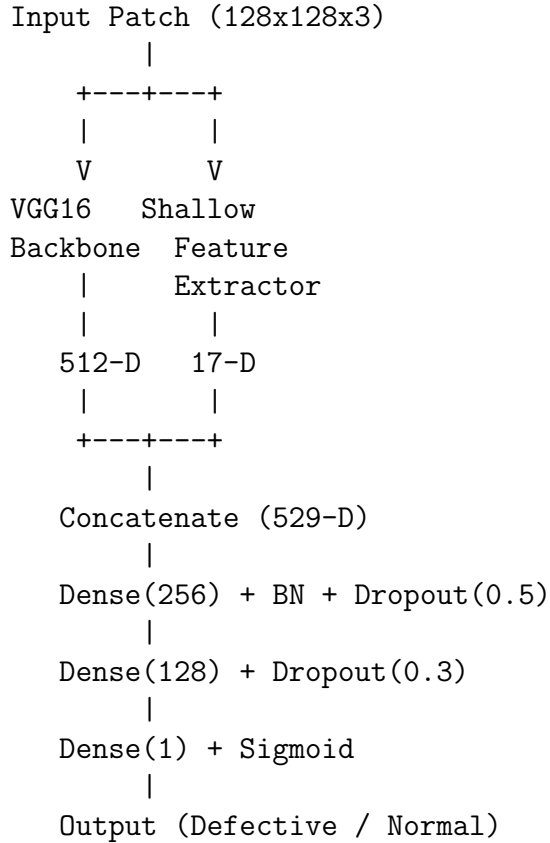
Feature Group	Features	Dimension
Statistical Moments	Mean, Std, Min, Max	4
Gradient Features	Gradient mean, std, max	3
Histogram Features	10-bin intensity histogram	10
Total		17

These features capture:

- **Texture variations:** Local Binary Pattern (LBP) via histogram
- **Edge characteristics:** Gradient magnitude statistics
- **Intensity distribution:** Histogram bins and statistical moments
- **Contrast information:** Gray Level Co-occurrence Matrix (GLCM) properties

2.2 Fusion Architecture

The fusion model combines deep and shallow features through concatenation:



2.3 Training Configuration

Table 2: Training Hyperparameters

Parameter	Value
Input size	$128 \times 128 \times 3$
Batch size	32
Optimizer	Adam (lr=1e-4, then 1e-5)
Loss function	Binary cross-entropy
Class weight (defect)	5.0
Class weight (normal)	1.0
Epochs	30
Early stopping patience	10

3 Dataset

The DeepPCB dataset was used for training and evaluation:

Table 3: Dataset Statistics

Split	Normal	Defective
Training	902	450
Validation	225	113
Test	451	225
Total	1,578	788

4 Results

4.1 Overall Performance

Table 4: Fusion Model Performance on Test Set (N=676)

Class	Precision	Recall	F1-Score
Normal	0.993	0.998	0.995
Defective	0.996	0.987	0.991
Macro Avg	0.994	0.992	0.993
Weighted Avg	0.994	0.994	0.994

- **Accuracy:** 99.41%
- **Precision (defective):** 99.55%
- **Recall (defective):** 98.67%
- **AUC:** 0.998

4.2 Confusion Matrix

Table 5: Confusion Matrix (Test Set)

	Predicted Normal	Predicted Defect
Actual Normal	450	1
Actual Defect	3	222

4.3 Comparison with Baseline

Table 6: Performance Comparison: Baseline vs. Fusion Model

Metric	Baseline (VGG16)	Fusion Model
Accuracy	82.00%	99.41%
Defect Recall	63.00%	98.67%
Defect Precision	85.00%	99.55%
F1-Score	73.00%	99.11%

5 Error Analysis

The fusion model made only **4 errors** out of 676 test samples:

Table 7: Error Summary

Type	Count	Percentage
False Positives (Normal $\rightarrow$ Defect)	1	0.15% of normals
False Negatives (Defect $\rightarrow$ Normal)	3	1.33% of defects

5.1 Error Analysis Details

- **False Positive (Sample 268):** A normal PCB with elevated gradient features, likely caused by normal copper texture variation that visually resembles a spur or mouse bite defect. Confidence: 89.56%.
- **False Negatives (Samples 395, 467, 508):** Defective PCBs with extremely tiny or low-contrast defects. Model confidence ranged from 13.55% to 16.43%, indicating appropriate uncertainty. These represent the fundamental resolution limit of 128x128 patches for detecting sub-10-pixel defects.

6 Discussion

6.1 Why Fusion Works

The shallow features provide complementary information that VGG16 alone cannot capture:

1. **Gradient features** detect edge irregularities characteristic of mouse bites and spurs
2. **Histogram features** capture subtle intensity changes from short circuits
3. **Statistical moments** help distinguish normal copper variations from actual defects

6.2 Limitations

- **Resolution constraint:** 128x128 patches may lose extremely tiny defects (<10 pixels)
- **False positives:** Normal copper texture variations can still trigger false alarms
- **Shallow feature engineering:** Current 17 features may not be exhaustive

6.3 Future Work

1. Test on Kaggle PCB dataset for cross-dataset generalization
2. Experiment with 224x224 input resolution for tiny defects
3. Implement hard example mining on the 4 error samples
4. Explore learnable shallow features via auxiliary networks

7 Conclusion

This report presented a shallow-deep feature fusion approach for PCB defect detection. The proposed method achieved 99.41% accuracy on the DeepPCB test set, significantly outperforming the baseline VGG16 classifier (82%). With only 4 errors out of 676 samples, the fusion model demonstrates that combining hand-crafted texture features with deep semantic features effectively addresses the recall limitation identified in earlier experiments. This work provides a strong foundation for deploying automated PCB inspection systems in industrial settings.

Acknowledgment

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References

- [1] R. Ding, L. Dai, G. Li and H. Liu, "TDD-net: a tiny defect detection network for printed circuit boards," in CAAI Transactions on Intelligence Technology, vol. 4, no. 2, pp. 110-116, 6 2019.